Math 257 Definitions and Theorems

Week 11: Modules 35-38

Fall 2023

35 Orthogonal Projections Onto a Line

Definition 35.1. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. The **orthogonal projection** of $\mathbf{v}$ onto the line spanned by $\mathbf{w}$ is

$$\text{proj}_{\mathbf{w}}\mathbf{v} := \frac{\mathbf{w} \cdot \mathbf{v}}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}.$$

Theorem 35.1. *Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. Then $\text{proj}_{\mathbf{w}}\mathbf{v}$ is the point in $\text{Span}(\mathbf{w})$ closest to $\mathbf{v}$; that is*

$$\text{dist}(\mathbf{v}, \text{proj}_{\mathbf{w}}\mathbf{v}) = \min_{\mathbf{u} \in \text{Span}(\mathbf{w})} \text{dist}(\mathbf{v}, \mathbf{u}).$$

Remark. Note that $\mathbf{v} - \text{proj}_{\mathbf{w}}\mathbf{v}$ (called the **error term**) is in $\text{Span}(\mathbf{w})^{\perp}$.

$$\mathbf{v} = \underbrace{\text{proj}_{\mathbf{w}}\mathbf{v}}_{\in \text{Span}(\mathbf{w})} + \underbrace{\mathbf{v} - \text{proj}_{\mathbf{w}}\mathbf{v}}_{\in \text{Span}(\mathbf{w})^{\perp}}$$

Theorem 35.2. *Let $\mathbf{w} \in \mathbb{R}^n$, then for all $\mathbf{v} \in \mathbb{R}^n$*

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \left(\frac{1}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}\mathbf{w}^T\right)\mathbf{v}.$$

Remark. Note that $\left(\frac{1}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}\mathbf{w}^T\right)$ is an $n \times n$ matrix, we call the **orthogonal projection matrix** onto $\text{Span}(\mathbf{w})$.

36 Orthogonal Projection Onto a Subspace

Theorem 36.1. *Let W be a subspace of $\mathbb{R}^n$ and $\mathbf{v} \in \mathbb{R}^n$. Then $\mathbf{v}$ can be written uniquely as*

$$\mathbf{v} = \underbrace{\hat{\mathbf{v}}}_{\text{in } W} + \underbrace{\mathbf{v}^{\perp}}_{\text{in } W^{\perp}}$$

Definition 36.1. We say $\hat{\mathbf{v}}$ is the **orthogonal projection** of $\mathbf{v}$ onto W - written $\text{proj}_W\mathbf{v}$.

Remark. If $(\mathbf{w}_1, \ldots, \mathbf{w}_m)$ is an ***orthogonal basis*** of W, then

$$\text{proj}_W\mathbf{v} = \left(\frac{\mathbf{v} \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1}\right)\mathbf{w}_1 + \ldots + \left(\frac{\mathbf{v} \cdot \mathbf{w}_m}{\mathbf{w}_m \cdot \mathbf{w}_m}\right)\mathbf{w}_m.$$

Theorem 36.2. *The projection map* $\text{proj}_W : \mathbb{R}^n \to \mathbb{R}^n$ *that sends* $\mathbf{v}$ *to* $\text{proj}_W \mathbf{v}$ *is linear.*

Definition 36.2. Let matrix P_W be the matrix $(\text{proj}_W)_{\mathcal{E}_n, \mathcal{E}_n}$ that represents proj_W with respect to the standard basis, we call P_W the **orthogonal projection matrix** onto W.

37 Least Squares Solutions

Goal: Suppose $A\mathbf{x} = \mathbf{b}$ is inconsistent. Can we still find something like a *best* solution?

Definition 37.1. Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. A **least squares solution** (short: LSQ solution) of the system $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

$$\text{dist}(A\hat{\mathbf{x}}, \mathbf{b}) = \min_{\mathbf{x} \in \mathbb{R}^n} \text{dist}(A\mathbf{x}, \mathbf{b}).$$

Theorem 37.1. *Let* A *be an* $m \times n$ *matrix and* $\mathbf{b} \in \mathbb{R}^m$. *Then* $\hat{\mathbf{x}}$ *is an LSQ solution to* $A\mathbf{x} = \mathbf{b}$ *if and only if* $A\hat{\mathbf{x}} = \text{proj}_{\text{Col}(A)} \mathbf{b}$.

Theorem 37.2. *Let* A *be an* $m \times n$ *matrix and* $\mathbf{b} \in \mathbb{R}^m$. *Then* $\hat{\mathbf{x}}$ *is an LSQ solution to* $A\mathbf{x} = \mathbf{b}$ *if and only if* $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$.

Theorem 37.3. *Let* A *be an* $m \times n$ *matrix with linearly independent columns and* $\mathbf{b} \in \mathbb{R}^m$. *Then*

$$\text{proj}_{\text{Col}(A)} \mathbf{b} = A(A^T A)^{-1} A^T \mathbf{b}.$$

38 Linear Regression

There are definitions or theorems in this module, see the lecture notes for examples.